THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT

KASULU TOWN COUNCIL

FORM FOUR PRE-MOCK EXAMINATION

CODE: 033 **BIOLOGY 1**

Time: 3:00 Hours **Wednesday, 21.05.2025**

INSTRUCTIONS

1. This paper consists of three sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B, and **two (2)** questions from section C.
3. All writing must be in **blue** or **black** pen, **except** for diagrams that must be drawn in pencil.
4. Communication devices and any unauthorized materials are **not** allowed in the examination room.
5. Write **Examination Number** on every page of your answer sheet(s).

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SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the following items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet.

(i) Form Three students took a thermometer and measure the temperature of the following organisms: **frog, tortoise, rat, lizard** and **rabbit** under cold and later under hot environment. Which organisms would the temperature remain constant?

A Rabbit, lizard and rat
B Rabbit, tortoise and rat
C Lizard rabbit and tortoise
D Frog, rabbit and tortoise
E Tortoise, lizard and frog

(ii) What is the aim of doing experiment in scientific investigation?

A Identifying a problem
B Finding solution to the problem
C Testing hypothesis
D Gathering information
E Recording results

(iii) Water from the roots flowering plants is transported up to the plants by different forces. Which of the following forces initiates and raises water to the least height?

A Root pressure
B Transpiration pull
C Cohesion force
D Adhesion force
E Capillarity

(iv) Which of the following is the function of hydrochloric acid in food testing experiments?

A To decolorize food sample
B To test reducing sugar
C To oxidize food sample
D To neutralize sugary food
E To hydrolyze complex to simple sugar

(v) Mr. John went to the hospital ask the treatment due to outward bending of his legs bones. This may be due to deficiency of:

A Vitamin A
B Vitamin C
C Vitamin B
D Vitamin K
E Vitamin D

(vi) Why skeletal muscle is usually found in opposing pairs?

A One muscle alone cannot extend a joint
B Muscle can only work by contraction
C One muscle alone cannot flex a joint
D Paired muscles are stronger
E Muscles normally work in antagonistic fashion

(vii) Mr. and Mrs. Msakatonge happened to have four children in their marriage. One of them was an albino. What is the possible genotype of Mr. and Mrs. Msakatonge?

A AA × Aa
B Aa × aa
C Aa × Aa
D aa × aa
E AA × Aa

(viii) Structure that controls the body balance in human is located in the:

A Outer ear B Inner membrane C Ear drum
D Middle ear E Inner ear

(ix) An organelle found on both plant cell and animal cell known as power house of the cell is;

A Vacuole B Cell wall C Cell membrane
D Mitochondria E Nucleus

(x) A doctor examines a patient complaining of difficult breathing. One potential cause identified is inflammation of the line, this could lead to:

A Increase air flow and lung capacity due to reduced friction.
B Difficult drawing air into the lungs due to narrowed airways.
C Improved gas exchange between air and blood due to increased surface area.
D Enhanced resistance to infections due to increased mucus production.
E The presence of cilia in the trachea which trap foreign particles.

2. Match the description of vitamin deficiency diseases in **List A** with their corresponding deficiency in **List B** by writing the letter of the correct response beside the item number.

List A	List B
(i) Rickets	A Cobalamine
(ii) Night blindness	B Quinnone
(iii) Scurvy	C Riboflavin
(iv) Excessive bleeding	D Calsiferol
(v) Beriberi	E Thiamine
(vi) Pellagra	F Ascorbic acid
	G Tocopherol
	H Niacin
	I Retinol

Section B (54 Marks)

Answer **all** questions from this section

3. (a) Draw a diagram of an animal cell and label the parts which perform the following functions:
   (i) Site for energy production.
   (ii) Control all the functions of the cell.
   (iii) Where organelles are suspended.
   (iv) Allow passage of some substances.

   (b) List three other functions of the part that perform the function in (ii) above.

4. (a) John attempted to observe plant cells using a prepared slide under a light microscope, but he was unable to obtain a clear image due to some challenges. Educate John on how to address the following challenges faced in order to obtain a clear image of the plant cell.
   (i) The slide containing specimen did not settle on the stage for observation
   (ii) He was unable to achieve sharp focus
   (iii) The image was too bright making it difficult to see the detail of specimen.

(b) Give three negative effects of disease.

(c) In a biology class your fellow students classified pigeon and butterfly in the same group based on the presence of wings. Explain three shortcomings of their classification.

5. An agricultural officer advised farmers to apply fertilizer that contain nitrogen, phosphorus and potassium in their farms with mineral deficiency. Which signs were shown by the plants that made the agricultural officer give such advice to farmers? Give three signs for each mineral deficiency.

6. (a) One of the most common accidents that occur at home and at school is falling, as a Form Four student explain three safety measures that can be considered in order to prevent this accident.

(b) Based on the theories of organic evolution describe the development of;
   (i) Webbed-feet in ducks.
   (ii) Long necks in giraffes.

7. (a) Give the meaning of the following biological terms;
   (i) Biotic factor.
   (ii) Abiotic factor.
   (iii) Food chain.
   (iv) Food web.

(b) Briefly explain five things that distinguishes biology laboratory from other school facilities.

8. Anna visited a food seller and bought the following foods; eggs, Irish potatoes and sunflower seeds.
   (a) Identify the type of nutrients found in each of the item listed.
   (b) Give four functions of the nutrients identified in the eggs.
   (c) What is the nutritional disorder caused by deficiency of the nutrients identified in eggs?
   (d) What are the two symptoms of the disorder identified in (c) above?

SECTION C (30 Marks)

Answer **two (2)** questions from this section

9. Discuss six significances of client-initiated counselling and testing in controlling HIV and other STIs.

10. The blood circulation is important for survival of human being. In six points justify this statement.

11. Life skills is one of the most important ideas that is required by the adolescent to cope with sexuality and sexual behavior but also enable them to deal effectively with everyday challenges. With six points discuss necessary life skills needed by the adolescent.